

Supplementary document

Come on, we dive directly into the heart of the subject with a frankly fascinating story. Welcome to this new decryption. It all begins with a little mental journey to a flourishing but totally fictional kingdom, which will be called Eteria. Let's imagine this world for a moment. There, the craftsmen were of incredible skill. They had made magic wagons, vehicles capable of swallowing miles without any horse. For the inhabitants of Eteria, it was an absolute revolution. But as often, there was a shock. And not least. To advance these wonders, each traveller had to shield his chest with what they called firestones. Extremely dense and especially heavy rock blocks. The logic was relentless. The longer the journey was, the more stones had to be cut off, and the cart was exponentially heavy. In the end, the vehicle bore its own burden. This is exactly the image, fairy tale version, of our very first energy storage system, and of what still limits us today. And that's where a certain Vera, a rather visionary alchemist, comes into play. The powerful guild of the merchants mocked her for years. But Vera asked THE question that was going to push everything. Why carry both the spark and the air that it burned, while the wind surrounds us all around? And when you think about it, it's just brilliant. Why burden yourself with a dead weight to create a reaction, while the ambient air is full of energy? This simple reflection was the starting point for a whole new approach. Well, actually, it didn't happen on its own. Vera's first attempts were, let's say, a disaster. By simply opening the windows of the reservoir to capture the wind, the reaction was unheard of violence. The foyer was suffocating, and it was creating species of sharp tips, dreadful ice thorns. These things grew, grew and ended up literally piercing the metal of the cart, ruining the entire system. Breathing the machine was far from being won in advance. But ingenuity always ends up paying. Vera hung on and eventually conceived a magic membrane. Concretely, a porous wall and above all ultra selective. His job? Let in the wind's breath, while ruthlessly blocking any impurity. The result was bluffing. No more stones in the trunk. The trolley was almost empty, feeding on the ambient air as it moved. The carriage was able to make four times the usual distance in a single journey. A real good one. Come on, we're leaving the kingdom of Etheria to land in an equally exciting world. That of modern electrochemistry. Because yes, this little story wasn't just for the pleasure of telling a story. It is actually a very precise metaphor, ideal for the ecologist or simply curious minds. We can decode all this very easily. The famous ultra heavy firestones. These are our current lithium-ion batteries, filled with heavy metals like cobalt or nickel. The breath of the wind is stupidly the oxygen of our ambient air. The terrifying ice thorns in pure and hard chemistry are called lithium dendrites. Parasite outgrowths caused by moisture and CO₂ that short-circuit the entire system. And finally, Vera's magic membrane, it's the equivalent of a solid electrolyte, able to filter the air to keep only vital oxygen, while blocking destructive water. So we come to the heart of the subject, the lithium-air or lithium-oxygen battery. Simply put, it is an accumulator that generates its current thanks to the oxidation of metallic lithium on one side and the reduction of oxygen in the ambient air on the other. The connection with the magic cart is direct, there is a battery that breathes the air around it to produce electricity. Section 2, theoretical potential, unmatched density. If the best labs in the world are focused on this, it is because lithium-air is simply the absolute grail of energy storage. The numbers literally give vertigo. To get an idea, our classic lithium-ion batteries are now capping around 250 Wh per kilo. The theory behind lithium-air, it, taps into 12,000 Wh per kilo. The gap is monumental. We are talking about an energy density that squarely rivals that of a full of gasoline. On paper, that means an electric car could drive 1,600 km without ever having to stop to recharge. Bye bye the anxiety of the power outage. The most beautiful is that the chemical mechanics behind is of great elegance, in two simple steps. At the discharge, lithium drops an electron, passes through our famous electrolyte and joins with the oxygen of the air to form lithium peroxide. And boom! Massive energy release. Then, for charging, we reverse everything. We send power, it breaks the peroxide, oxygen gently returns to the atmosphere and lithium returns to pure metal, fresh for the next trip. Section 3, the limits of reality, the chemical challenge. Well, it's time to come down a little late. The history of the terrias made it seem that with the right membrane, the trick was played. But the ruthless laws of physics are a little more stubborn than that. There are so many obstacles that still prevent us from applying all this. First, energy efficiency is quite bad. In other words, a lot more juice is needed to recharge it than it gives back to use. The difference, she's going wild in heat. Next, degradation. Internal chemistry is so reactive that the battery is used at high V speed. We are far from the thousands of cycles of our current batteries. And then, the big black dot, its almost unhealthy sensitivity to the humidity and CO₂ of the ambient air. For this to work, we need super complex filtration engineering. Section 4, the future of lithium-air beyond 2030. So, is this a lost cause? Absolutely not. Real titans from industry, like the CATL battery giant, put the package in research and development to blow up these technological locks. And we only have what to hope for. Let's keep that figure. 1200. Recently, laboratory prototypes have managed to

achieve an impressive density of 1200 Wh per kilo, while maintaining the road for more than 1000 load cycles. And this at room temperature. It's huge. This is made possible by new solid ceramic electrolytes. And it completely pollutes what we do best today in lithium-ion. I mean, the question that burns our lips is when do we get this in our cars? Well, you'll have to be patient. To succeed in such a prowess in a highly controlled lab is beautiful. But manufacturing millions of these ultra-sensitive units on a large industrial scale is another challenge. That's why experts rely carefully on an arrival after 2030. The transition to mass production still requires enormous innovations. To close this decryption, let us keep this reflection on our heads. The future of our energy will really depend on this incredible ability to transform the magic of the ambient air, as in the account of Etheria, into a real solid industrial technology. We must cross heavy metals dead weight and finally let our energy reserves breathe. This is probably the most exciting scientific challenge of this century. If we blow these last locks, we're not just talking about cars driving further, we're talking about a total overhaul of the energy paradigm of our planet. So is the world ready to capture the breath of the wind to reinvent its future?